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Systematically Improving RAG Applications

Week 1: Introduction

Jason Liu and Dan Becker

Thank you for trusting us

Agenda

- Who are we?
- Who are you?
- What does success look like?
- The RAG playbook
- Q&A

Who am I?

Jason Liu

Data scientist

Meta

- Graph and content analysis to identify cyber crime
- "RAG apps to fight internet evil"

Staff MLE

STITCH FIX

- Creation of robust recommendation framework and observability tools, handing 350M+ recommendations per week.
- In 2017-2018, implemented vision, multimodal search, and VAE-GANs (boosting revenue by $50M+)
- Oversaw a $400k budget for data curation
- Implemented upstream search systems for similar items, complementary items, outfits, curated collections

Applied AI Consultant

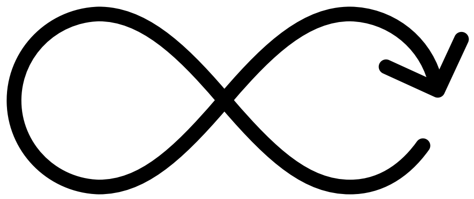
- Strategic consulting to solve problems related to RAG, query understanding, prompt engineering, embedding finetuning, MLOps observability
- Work with leading companies on AI products such as Limitless AI, Sandbar, Raycast, Tensorlake, Dunbar, Bytebot, Naro, Trunk Tools, New Computer, Kay.ai, Modal Labs, Pydantic, and Weights & Biases.

Who am I?

Why don’t you build stuff rather than be a consultant?”

- Hand injury prevents me from coding too much

My highest leverage work now is in advising and education



The focus on my career:

- In 2018: “Make vision work”
- In 2022: “Make text work”

I have 6+ years of experience:

- Building visual product search with embeddings
- Building recommendation systems
- Improving ML-based product reliability

I want to share what I know…

Who am I?

Dan Becker

Data Scientist & Head of Product

- Finished 2nd (out of 1350 teams) in Kaggle competition with $500k prize
- Led product team as early employee at DataRobot during period of 400% YoY growth
- Designed 1st commercial ML drift detection platform

Data Science Educator

- Led creation of AI education platform used by 45k MAU

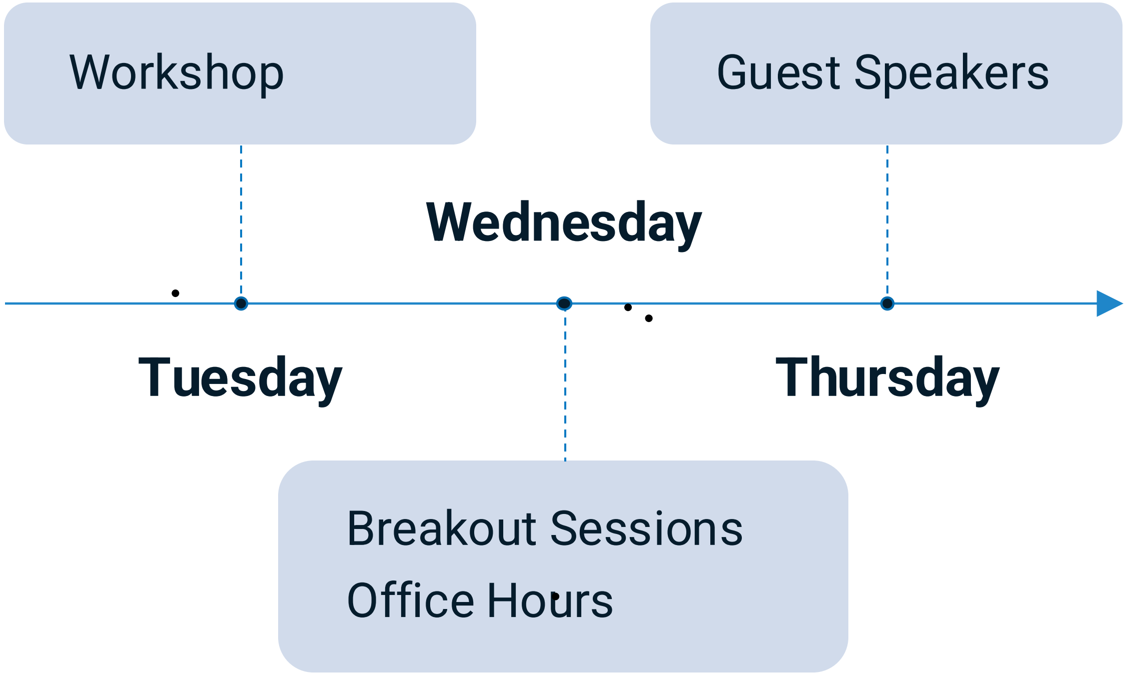
Founder

- Founded software platform to combine ML and analytic models into simulations for algorithmic decision optimization
- Acquired by DataRobot, where I led AI Development Tools product team (~40 engineers)

Applied AI Consultant

- Working with clients on 3 continents
- Advised ~15 projects across RAG, fine-tuning, extraction from multimodal data sources, etc.

Course Structure



Week 1: Intro to the Playbook

Week 2: Identifying Areas of Improvement

Week 3: Adding Retrieval Methods + Non-Text Data

Week 4: Routing Queries

Week 5: Improving Representations

Week 6: Product Design and Prompting

Agenda

Who are we?

Who are you?

What does success look like?

The RAG playbook

Q&A

Who are you?

We're still learning who you are, week 1 will be a bit slow as we calibrate to everyone's level

36%

Co-founders, Executives

40%

SWE, MLE, Data Scientist

Agenda

Who are we?

Who are you?

What does success look like?

Build a “system” to improve RAG

The RAG playbook

Q&A

What is a system?

A structured approach to solving problems that guide how we think about and tackle challenges

In the context of building an AI product:

1. A **framework for evaluating** different technologies and tools
2. A **decision-making process** for prioritizing development efforts
3. A **methodology for diagnosing and improving** application performance
4. A **set of standardized metrics** and benchmarks to measure success

Why do we need a system?

Systems free up mental energy for innovation and for tacking unique challenges

- Companies often waste time guessing instead of testing new technologies.
- Quantitative results and qualitative assessments are both necessary.
- Securing resources for developing new capabilities is challenging without being data-driven.

What does success look like?

A Lower anxiety when told “make the AI better”

B Feel less overwhelmed when told “look at the data”, instead leverage data to:

- Identify high-impact tasks
- Prioritize effectively to make informed tradeoffs
- Choose relevant metrics that correlate with business outcomes

C Drive better outcomes:

- Higher user satisfaction
- Higher retention
- Higher and more frequent usage

Where have I done this?

- **Personal agents**
- **Construction**
- **Workflow automations**
- **Sales and marketing**
- **Transcript and data mining**
- **Private equity / due diligence**

What does the path to success look like?

A I focus on **experimentation** rather than a vague "make the AI better"

B When I hear "look at the data", I ask

- **Why** am I looking at my data?
- **What** am I looking for?
- **When** I see the signals:
    - **What** can I do with this information to improve my application?
    - **Is the juice worth the squeeze?**

C Once the flywheel is in place, success is defined by **consistency - doing the most obvious thing over and over**

- This will be boring… because a better app is a result, not an effort
- **Your only job is to apply consistent effort**
- For example, if you want to build muscle, you track calories and workouts (which is boring)

Agenda

RAG is more than just R-A-G

What you think RAG is:

What RAG actually is:

Good Retrieval in RAG resembles a four-step recommendation model

Engineers focus on generation without even knowing if the right information is retrieved

Only through improved search can you improve retrieval which will ultimately improve generation

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The LLM chooses from different search backends based on the query

Retrieval is not only multi-step, but also multi-index

Instead of implementing R-A-G, implement RecSys in an LLM sandwich

Step 1:

- Chat interface or web UI intakes original query
- Prepare a list of queries that are routed to specific search engines
  - Planning, CoT, Parallel Tool use

Step 2:

- Based on original routing steps, retrieval, filtering, scoring, and ordering is done

Step 3:

- Produce representations for the chunks
- Choose a generation prompt to reply to the LLM's

Step 4:

- Build a special UI to render responses beyond text streams
- For example: profile cards, pdf viewers, citations, feedback buttons

Example

instructor - example.py

```python
def rag_app(question, messages, k):
    searches, qualified_query = query_understanding(question, messages)
    answer_prompt = answer_prompt(qualified_query)

    results = [search.execute() for search in searches]
    results = order_results(results)[:k]
    results = format_chunks(results)

    answer = answer_question(question, results, answer_prompt)
    return answer
```

The RAG Flywheel

Thesis: The principles we've applied in search are highly relevant to what we want to do with RAG

Step	Description
❶ **Initial implementation**	Start with a basic RAG system setup
❷ **Synthetic data generation**	Create synthetic questions to test the system's retrieval abilities
❸ **Fast evaluations**	Conduct quick, unit test-like evaluations to assess basic retrieval capabilities (e.g., precision, recall, mean reciprocal rank), and explain why each matters
❹ **Real-world data collection**	Gather real user queries and interactions. Ensure feedback is aligned with business outcomes or correlated with important qualities that predict customer satisfaction
❺ **Classification and analysis**	Categorize and analyze user questions to identify patterns and gaps
❻ **System improvements**	Based on analysis, make targeted improvements to the system
❼ **Production monitoring**	Implement ongoing monitoring to track system performance
❽ **User feedback integration**	Continuously incorporate user feedback into the system

The RAG Playbook process

Step 1: Address cold start

a. Generate synthetic Data
b. Establish general baselines for:
   i. Recall
   ii. Precision
   iii. Lexical search
   iv. Semantic search
   v. Re-rankers
c. Use this to test all other hypotheses

Step 2: Instrument and observe user data to identify strengths and weaknesses in clusters

a. Topics and capabilities identified with domain experts
b. Understand and underperforming query segments
c. Convert offline analysis to online classifiers and query routers to monitor production data

Step 3: Implement new routers, tools, and capabilities

a. Evaluate experiments against both offline and product metrics
b. Explore and Exploit given distributions of capabilities
c. Iterate and monitor continuously

Rinse and repeat:

a. Continuously conduct exploratory data analysis, partner with users and domain experts, and extend capabilities
b. Understand the process of evaluating candidate selection
c. Use topic modeling,

The RAG Playbook

Thesis: The principles we've applied in search are highly relevant to what we want to do with RAG

Week 1

1. **Initial implementation**
2. **Synthetic data generation**
3. **Fast evaluations**

Week 2

4. **Real-world data collection**
5. **Classification and analysis**
6. **System improvements**
7. **Production monitoring**
8. **User feedback integration**

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